

SightOps

An Agentic Visual Reliability Engineer

Core idea

SightOps turns an ordinary camera into an active inspection agent. OpenCV 5 extracts measurable visual evidence from equipment; when evidence is uncertain, the agent requests or triggers a better view, re-inspects, reasons over the new evidence, and takes a controlled operational action.

Team	Yabloko Labs
Team representative	Santhosh Balasa - Founder & CEO, Yabloko Labs
Featured paths	Agentic Vision + Best Use of COOL
Primary domain	Industrial inspection / Physical AI / Reliability operations
Core stack	OpenCV 5, COOL, AWS Graviton, Amazon Bedrock, S3, DynamoDB, Lambda, CloudWatch

One-line pitch: SightOps does not merely recognize what a camera sees. It investigates physical equipment: observe, measure, identify uncertainty, acquire better evidence, decide, and act with human control where required.

1. Problem & Real-World Impact

Visual inspections remain common across factories, data centres, warehouses, utilities, building systems, solar installations and telecom infrastructure. Operators often need to inspect indicator lights, analogue gauges, switch positions, leaks, obstructions or other visible state before deciding whether equipment is healthy. Manual inspection is repetitive, difficult to scale and dependent on staff being physically available.

A conventional camera analytics system can detect predefined objects, while a general-purpose vision-language model can describe a frame. Neither approach inherently performs an investigation. A useful reliability system must know when the current evidence is insufficient, obtain better evidence, correlate multiple observations, and choose the next safe action.

SightOps addresses that gap with active perception.

The system uses OpenCV 5 as the measurement layer. It detects and rectifies equipment regions, reads visual state, tracks changes and estimates confidence. Those structured observations become inputs to an agent. Low-confidence evidence changes the agent's next tool call: for example, it can request a closer view of a gauge or command a small pan/tilt camera to reposition. The new frame is analysed again before a diagnosis or operational action is proposed.

Intended impact

Reduce repetitive inspection effort, shorten time-to-detection for visible faults, create auditable visual evidence, and enable inexpensive camera-based monitoring without requiring a GPU at every inspection point.

Target users / beneficiaries

Industrial maintenance teams	Routine inspection of panels, gauges and status indicators.
Data-centre / facility operators	Remote visual checks and incident evidence.
Warehouse / utility operators	Continuous inspection of known physical assets.
SMEs	Low-cost monitoring using existing cameras instead of specialized vision hardware.

Why this is more than object detection

The key technical behaviour is a closed perception-decision-action loop. A visual result is not the endpoint. It can cause the system to change camera position, invoke a different OpenCV measurement, compare against historical evidence, escalate, create an incident, or ask a human to approve remediation.

2. OpenCV 5 & Agentic Vision Implementation

OpenCV 5 will perform substantive image/video analysis rather than serving only as an image loader. The initial demonstrator will use a mock equipment/control panel containing analogue gauge(s), coloured status indicators and switch/state markers so that every perception step can be reproduced and quantitatively evaluated.

OpenCV operation	Role in SightOps
Camera calibration	Correct lens distortion and establish consistent geometry.
Perspective correction	Rectify oblique views of panels and gauges.
ROI detection	Locate known inspection regions and isolate components.
Indicator-state analysis	HSV/Lab colour analysis, thresholding, morphology and contour filtering for LED/status state.
Analogue gauge reading	Circle/contour localisation, line/needle estimation and angle-to-value mapping.
Change detection	Compare current vs prior observations to trigger deeper inspection.
Image-quality scoring	Blur, exposure, occlusion and geometric quality checks used as confidence signals.
Frame selection / tracking	Select useful evidence and maintain component identity across frames.

Agent workflow

```
CAMERA / VIDEO
|
v
[ OpenCV 5 / COOL ] -- visual measurements + confidence --> [ INSPECTION AGENT ]
^ |
| |
+-- new view / new ROI / new measurement <--- PLAN -----+
|
+-----+
| | |
v v v
Re-inspect Create incident Human approval
|
v
Controlled action
```

Example trace: OpenCV detects a red warning indicator but estimates gauge-reading confidence at 0.41. The agent does not diagnose from weak evidence. It requests a closer gauge view. OpenCV re-runs perspective correction and gauge measurement, returning 87 PSI at 0.96 confidence. The agent correlates the high pressure with the warning state, creates a high-severity incident and requests human approval before a simulated shutdown.

3. Planned AWS Architecture

The project will run a meaningful vision workload on AWS and will intentionally target both featured paths. The core OpenCV/COOL workload will be deployed on an AWS Graviton Arm instance so the same reproducible image set can be benchmarked against an appropriate standard OpenCV baseline.

AWS component	Planned use
AWS Graviton + COOL	Core image-processing path: resize/rectification, thresholding, contours, quality checks and other supported operations.
Amazon Bedrock	Agent reasoning and, where useful, secondary VLM interpretation of selected evidence - not a replacement for OpenCV measurements.
Amazon S3	Store inspection frames, cropped evidence and reproducible evaluation inputs.
Amazon DynamoDB	Inspection state, component state, confidence, incident state and agent workflow metadata.
AWS Lambda	Event-driven operational actions such as incident creation, notification or simulated remediation.
Amazon CloudWatch	Latency, tool-call traces, failures, resource metrics and auditability.

High-level cloud flow

```
Camera / uploaded test stream
|
v
AWS Graviton Arm compute
[ OpenCV 5 + COOL ]
|
+---- evidence ----> Amazon S3
|
+---- state -----> DynamoDB
|
v
Agent / Bedrock
|
+---- next OpenCV tool / camera instruction
|
+---- Lambda action / human approval
|
v
CloudWatch trace
```

COOL benchmark objective

Run the claimed core image-processing workload on AWS Graviton with COOL and compare against a pinned standard OpenCV 5 baseline using identical inputs. Report latency, throughput, CPU utilization and estimated processing cost. No performance claim will be made until measurements are collected.

4. Evaluation, Judge Demo & Responsible Operation

Evaluation plan

Measure	Evidence
Visual measurement accuracy	Gauge absolute error; LED/state classification precision/recall; ROI localisation success.
Active-perception success	% of initially uncertain inspections resolved after one or more agent-directed observations.
Task success	% of scenarios reaching the correct final operational state/action.
Confidence behaviour	Calibration of low-confidence cases; false escalation and missed escalation rates.
Performance	End-to-end inspection latency; OpenCV/COOL operation latency; throughput and CPU utilization.
Reliability	Tool failures, retry success, degraded-mode behaviour and agent trace completeness.

Judge demonstration

The live demo will use a small physical mock control panel visible to a webcam. In the normal state, SightOps performs an inspection and records healthy measurements. A fault is then introduced: a warning indicator changes and a gauge moves outside its normal range. The first camera angle will intentionally make the gauge unreliable. The agent will observe low confidence, request or command a better view, re-run OpenCV analysis, obtain high-confidence evidence, diagnose the simulated fault, create an incident and request human approval for a simulated shutdown. The demonstration will expose the OpenCV measurements, confidence values and agent tool trace so judges can see exactly why each later action occurred.

Failure cases to test

Glare, low light, blur, partial occlusion, extreme camera angle, visually similar indicator colours, gauge needle ambiguity, camera movement, network interruption and unavailable model/tool calls. The system will prefer an explicit 'insufficient evidence' state over inventing a measurement.

Responsible operation & human control

SightOps is designed as decision support for inspection, not as an unrestricted controller of safety-critical machinery. High-impact remediation remains behind explicit human approval in the competition demonstrator. Every incident retains the relevant visual evidence, OpenCV measurements, confidence values and agent decisions. The project will use synthetic or team-owned demonstration imagery and avoid biometric identification or person surveillance.

Success criterion

A judge should be able to see a visual observation change the agent's next perception step and then see the improved evidence change the final operational action - with the entire chain reproducible and observable.

5. Featured Paths, Delivery Plan & Team

Featured paths

Agentic Vision	YES	OpenCV measurements and confidence directly alter later plans, tool calls, camera/view request
Best Use of COOL	YES	Core supported vision operations will execute on AWS Graviton using COOL, with a reproducible

Build plan

Phase 1	Reproducible mock panel, camera pipeline, OpenCV 5 baseline, labelled evaluation frames.
Phase 2	Gauge/indicator/quality tools, confidence outputs, state model and automated tests.
Phase 3	Agent orchestration with active-perception loop and human-approval boundary.
Phase 4	AWS deployment, S3/DynamoDB/Lambda/CloudWatch integration and COOL on Graviton.
Phase 5	Benchmarking, failure-case evaluation, technical report and <=5 minute judge video.

Team bio

Yabloko Labs builds agentic AI software. The project is led by **Santhosh Balasa**, Founder & CEO, a software and AI engineer with experience across Python, Rust, C/C++, cloud platforms, backend systems, machine learning infrastructure and agentic AI. His engineering background includes production work involving AWS, Azure and GCP, FastAPI and distributed backend systems, ML/AI workloads, data platforms and performance-focused systems engineering. Recent independent builds have focused on autonomous agents, tool orchestration, retrieval/memory systems and human-in-the-loop workflows.

The team will apply that agent-infrastructure experience to a computer-vision problem where OpenCV is not an accessory: it is the evidence-producing tool layer that controls what the agent can safely infer and what it chooses to do next.

Why SightOps

The competition asks for vision systems that see, reason and act. SightOps is designed around exactly that loop: OpenCV 5 sees and measures; the agent reasons about evidence quality and system state; AWS executes and records the resulting inspection workflow; human approval constrains consequential actions.

Project: SightOps - An Agentic Visual Reliability Engineer

Team: Yabloko Labs

Representative: Santhosh Balasa

Competition: OpenCV AI Competition 2026, powered by AWS